Ali Saeed Chaudhary

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EDUCATION

Iowa State University Ames, Iowa

B.S., Mechanical Engineering. Cyber-Physical Systems Minor. ***GPA: 4.0.*** *May 2028*

Korea University Seoul, South Korea

Study Abroad Program. *June 2026*

Relevant Coursework: Studied fundamentals of robotics and mechatronics, Developed reinforcement learning algorithms using PPO, TRPO, and VLA Models for robotics applications, Gained coursework knowledge in nuclear power and nuclear energy

EXPERIENCE

Research Assistant Aug 2026 – Present

Iowa State University, Coordinated Systems Lab *Ames, Iowa*

- Partnered with RTX Corporation to develop an air traffic prediction network
- Researching airspace trajectory prediction with cascading confidence intervals using RNNs
- Writing code for neural network models and conducting research into available training data; building datasets and collecting relevant literature in the project's preliminary phase

Radiology IT Support Coordinator Nov 2024 – Feb 2026

Concord Medical Management, LLC *Flower Mound, Texas (remote)*

- Coordinated live data flow between remote health facilities and radiologists, resolving technical issues under tight deadlines
- Optimized system escalation workflows and communications to ensure timely medical IT support

ACTIVITIES & LEADERSHIP

Cardinal Space Mining, NASA Lunabotics Aug 2026 - Present

Collections Sub-Team Member *Ames, Iowa*

- Leading dust-cover design for the collections sub-system of a lunar regolith mining rover; focused on improving robustness over the prior year's design while maintaining functionality and manufacturability

Cyclone Rocketry, sponsored by Boeing Oct 2024 - Dec 2025

Mechanical Sub-Team Member *Ames, Iowa*

- Designed and developed a model rocket; proposed and lead the designing of an airbrakes flight mechanics system; modeled components in SolidWorks and ran FEA simulations in Ansys

PROJECTS

Gravity-Powered Irrigation System Aug 2025 – Dec 2025

- Designed, planned, and built a portable gravity-powered irrigation device for ME 2700
- Served as project manager, allocating roles across the team
- Led design of the gearbox and primary mechanism; simulated system in SolidWorks and hand-manufactured from wood

The Euclid Project Jan 2023 – July 2023

- Built an NLP neural network in Python and spaCy to detect and rate offensive or dangerous messages on online forums
- Implemented supporting infrastructure including sockets and servers to link separate system components
- Deployed as a moderation tool in a large online forum with over 20,000 members for live testing and data collection

Private Home Server June 2026 – Present

- Building a private media server for storing, hosting, and distributing services and files
- Deployed on Debian, accessed remotely via TailScale; gaining hands-on experience in network systems management

SKILLS

CAD & Simulation: SolidWorks, Ansys FEA, CAM

Programming & ML: Python, spaCy, RNNs, PPO, TRPO, VLA Models, Robotics

Systems & Tools: Debian Linux, TailScale, Network Systems Management

Manufacturing: CNC Milling, Welding, Metal Working, 3D-Printing